

Aetiology

The lesions of molluscum contagiosum are a benign skin eruption caused by infection with a large DNA virus of the Poxviridae family, Molluscipox genus.

The molluscum contagiosum virus (MCV) has two main subtypes—types 1 and 2—which account for virtually all infections, although genotypic analysis has identified other rarer subtypes. Type 1 is the most prevalent, with variable distribution across different geographical regions. Two main clinical presentations are observed:

- (i) Lesions on the face, neck, trunk, and arms, predominantly seen in children.
- (ii) Genital, pubic, lower abdominal, upper thigh, and/or buttock lesions, typically acquired through sexual contact and commonly seen in young adults.

Lesion appearance does not differ between subtypes, and individual infections are usually caused by a single subtype (though dual infections have been reported in persons living with HIV). There is no significant difference in the anatomical distribution of subtypes 1 and 2; however, MCV subtype 2 is slightly more common in genital lesions and is also more frequently associated with immunosuppression and HIV infection, where the disease may be more extensive and severe. Subclinical infection is common, and seroconversion (appearance of antibodies in blood) does not always occur, even in symptomatic cases.

Transmission

Transmission occurs primarily through direct physical contact. Most evidence comes from cohort studies in children, but transmission has also been documented in contact sports. Increased transmission has been linked to swimming and co-bathing, as well as fomite spread via shared towels or sponges. Congenital transmission has been reported in rare cases. In adults, the

most common mode of transmission is direct skin-to-skin contact during sexual intercourse.

Clinical Findings

Sexually transmitted molluscum contagiosum typically affects the anogenital area, including the external genital organs, inguinal folds, inner thighs, and suprapubic region. Less commonly involved sites include the areola and nipple, cervix, oral mucosa, and the palms and soles.

The incubation period typically ranges from 2 to 7 weeks but can extend up to 6 months.

Characteristic lesions, known as "mollusca," are dome-shaped, smooth, pearly, firm papules that are skin-colored, pink, yellow, or white, measuring 2–5 mm in diameter, with central umbilication. Squeezing a lesion releases a cheesy material composed of degenerated keratinocytes and viral particles.

Dermoscopy can aid diagnosis by revealing a central polylobular white-yellow structureless area surrounded by blood vessels arranged in a crown-like pattern.

In immunocompetent adults, the number of lesions usually ranges from 1 to 30. Lesions may appear in clusters (agminated form) or in linear arrangements due to autoinoculation, reflecting the isomorphic (pseudo-Koebner) phenomenon.

Molluscum contagiosum is typically asymptomatic. However, local pruritus or discomfort may occur, increasing the risk of self-inoculation to other skin areas.

Atypical presentations include: - Giant lesions - Cystic forms - Ulcerated lesions - Follicular involvement - Lesions resembling condyloma acuminatum or sebaceous nevus - Pyogenic granuloma-like lesions - Pseudolymphomatous appearance - Cellulitis- or abscess-like presentations

An eczematous reaction known as molluscum dermatitis—characterized by erythema, scaling, and inflammation—may develop around lesions due to a local immune response and often precedes clinical resolution. Conversely,

molluscum contagiosum occurring in the context of atopic dermatitis (termed eczema molluscatum) can present a clinical challenge, manifesting as widespread, inflamed lesions with scaling or hemorrhagic crusts and significant pruritus.